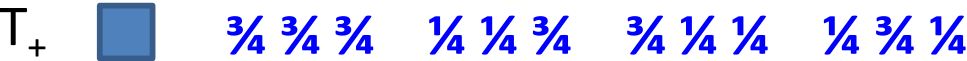
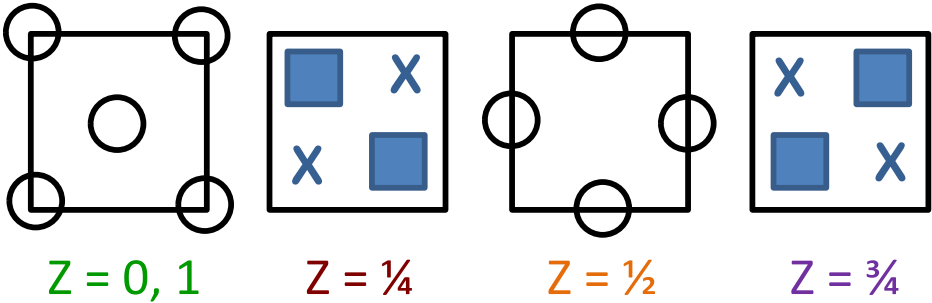
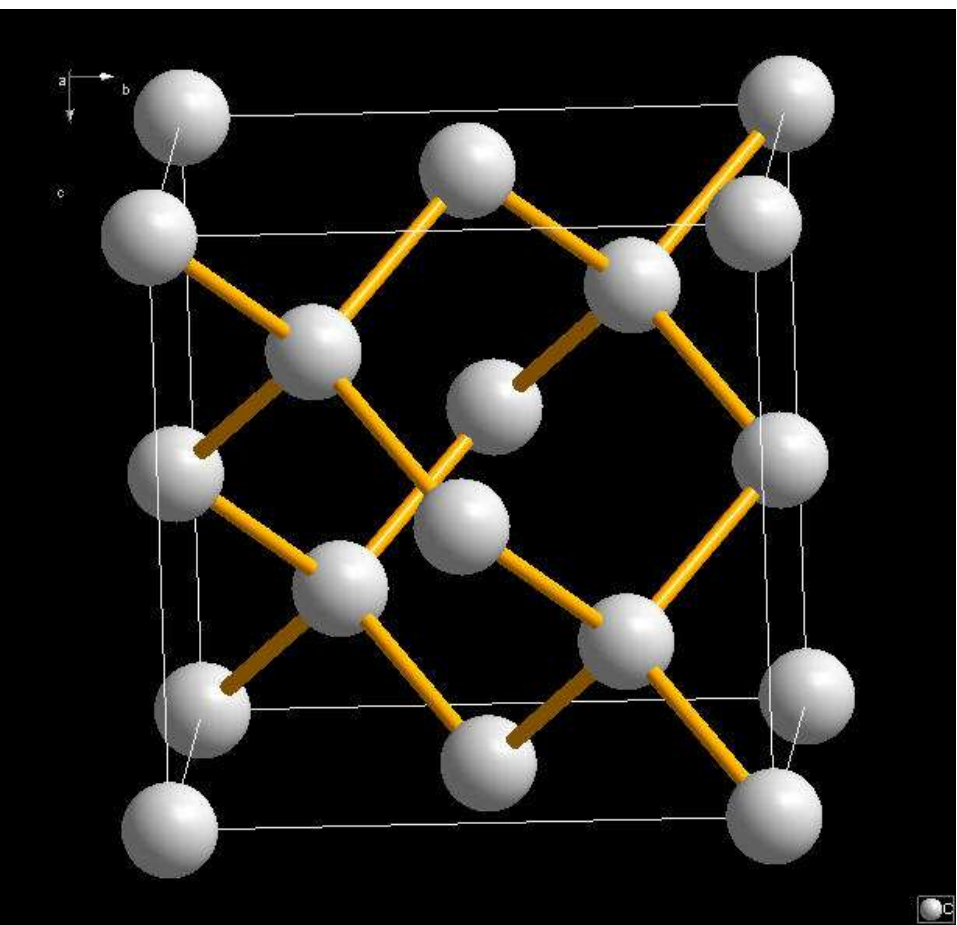


Special structures

Diamond



Covalent type structure

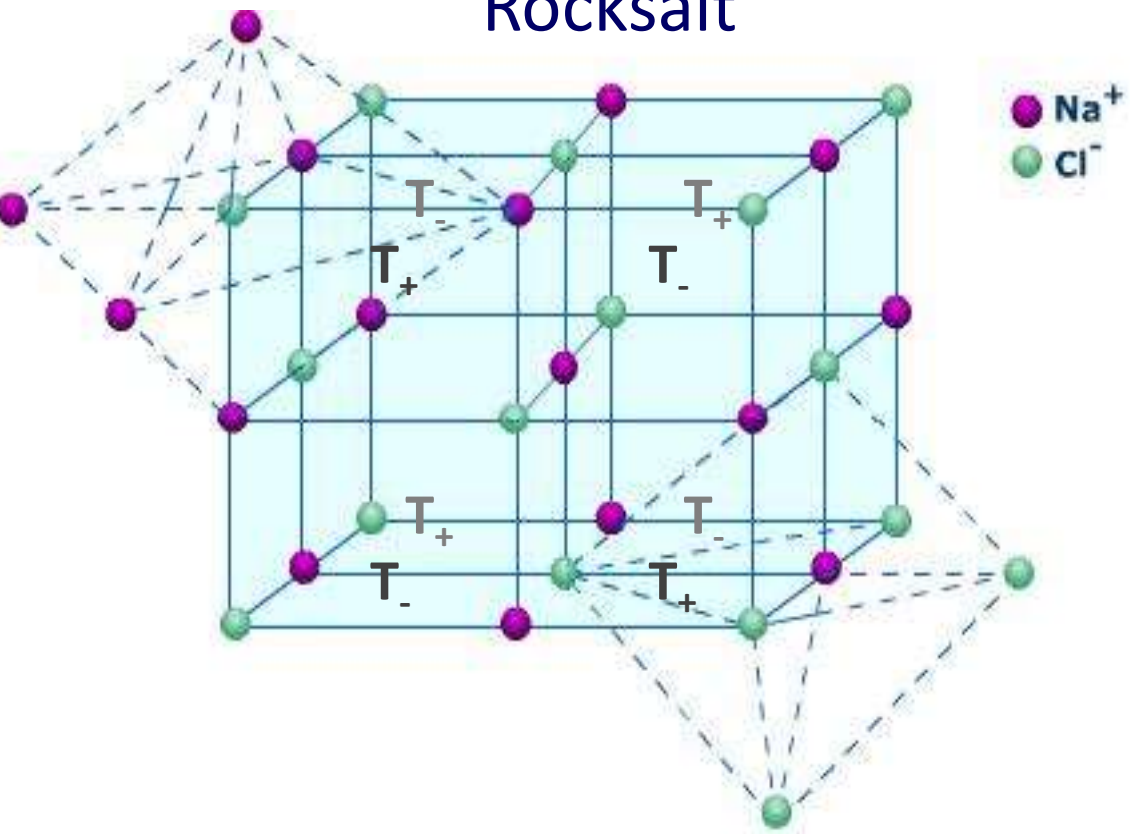
Alternate td voids are occupied by carbon atoms

Normal FCC structure: $\sqrt{2} a = 4 r$
 $\therefore a = \sqrt{8} r$
 $r = \text{radius of one atom}$

Diamond structure: $\sqrt{3} a = 8 r$
 $\therefore a = \sqrt{\frac{8}{3}} \times \sqrt{8} r$
 $a \text{ increases by } 1.633 \text{ \AA}$

Rocksalt or Zinc blende or Antifluorite

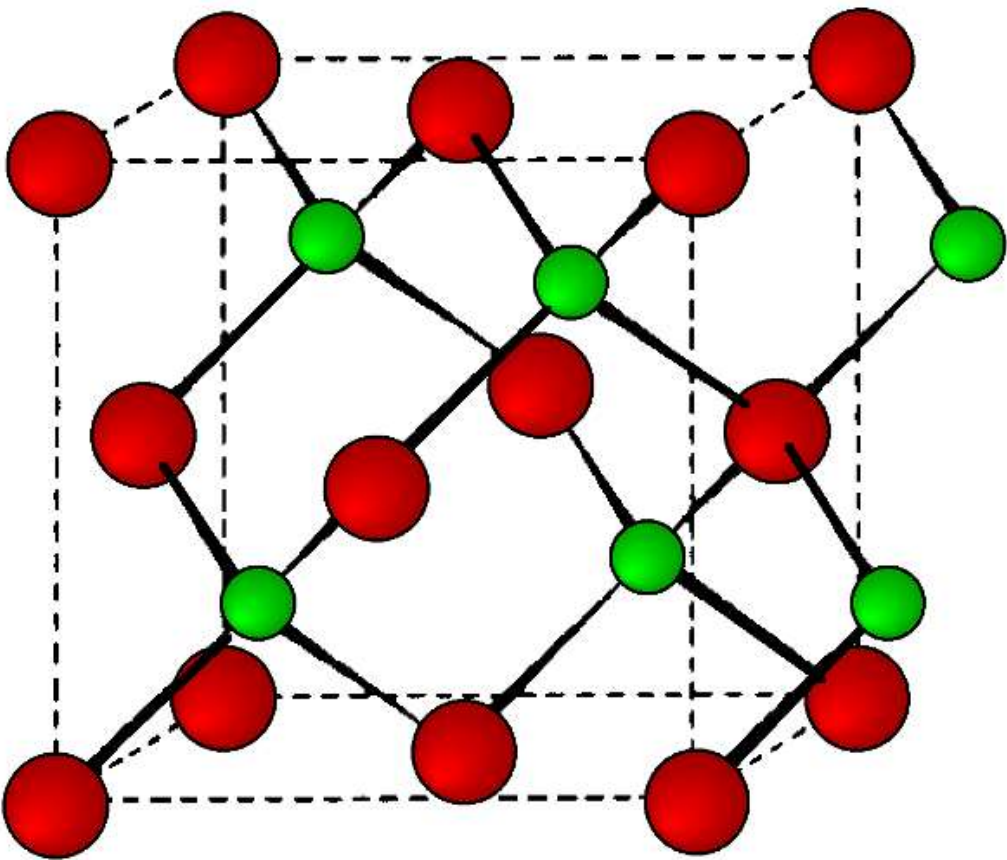
Rocksalt



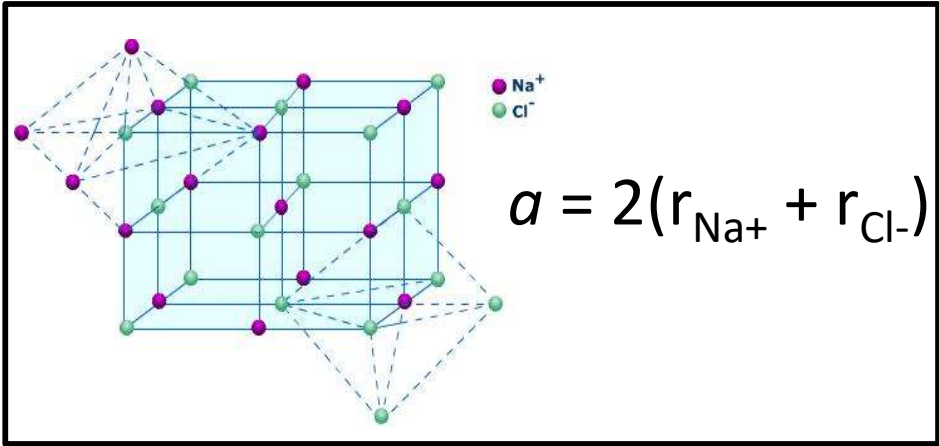
Oh	$\frac{1}{2}$	0	0	0	$\frac{1}{2}$	0	0	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
T ₊	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{1}{4}$
T ₋	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{4}$

- Rocksalt** (NaCl type) : Oh sites occupied by cations, T₊, T₋ empty
- Zinc blende** (ZnS type) : T₊ (or T₋) occupied, Oh, T₋ (or T₊) empty
- Antifluorite** (Na₂O type) : T₊ & T₋ occupied, Oh empty

Zinc Blende



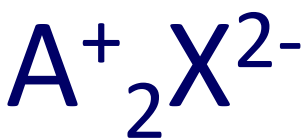
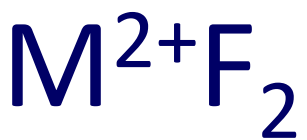
$$\frac{\sqrt{3}a}{2} = 2r_{S^{2-}} + 2r_{Zn^{2+}}$$



Zinc blende (ZnS type) : T₊ (or T₋) occupied, Oh, T₋ (or T₊) empty

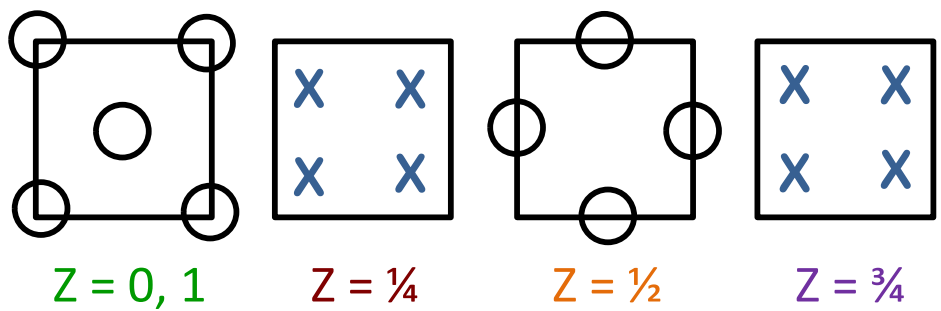
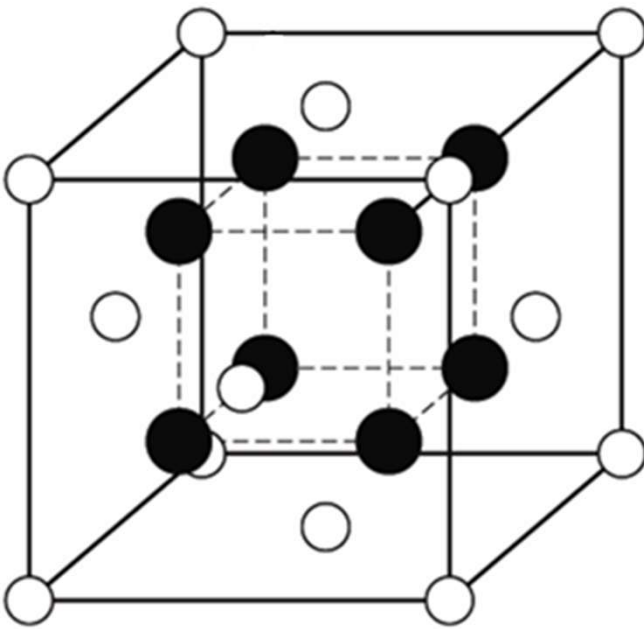
CuS & CdS crystallize in this structure B3 type

Fluorite & Antifluorite

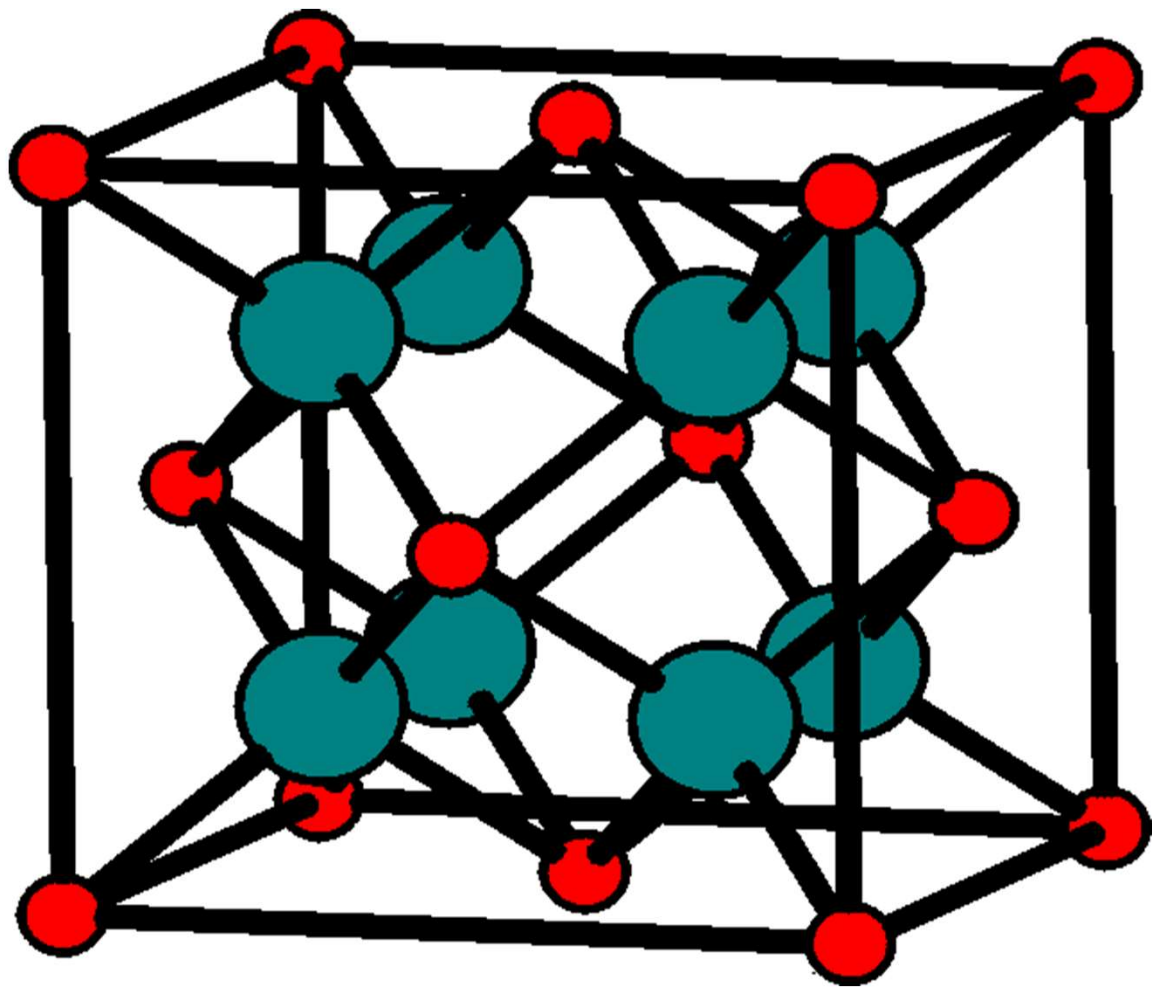
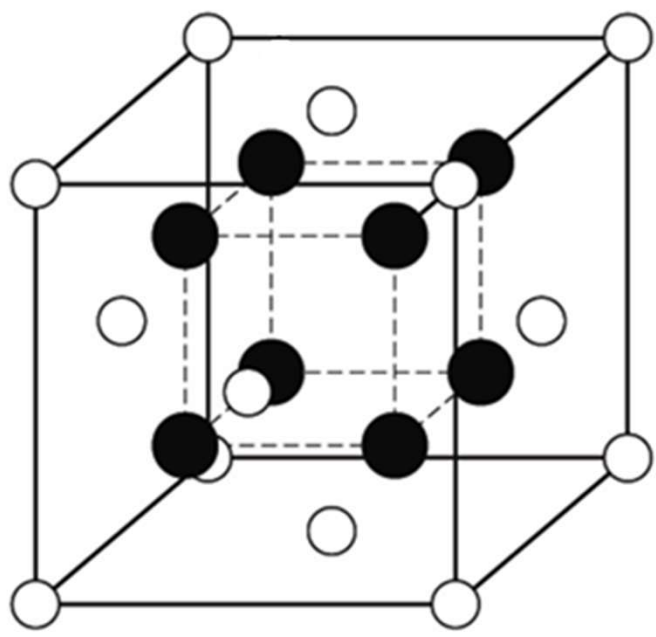


- Cations → 8 coordinate
- × Anions → 4 coordinate

- × Cations → 4 coordinate
- Anions → 8 coordinate

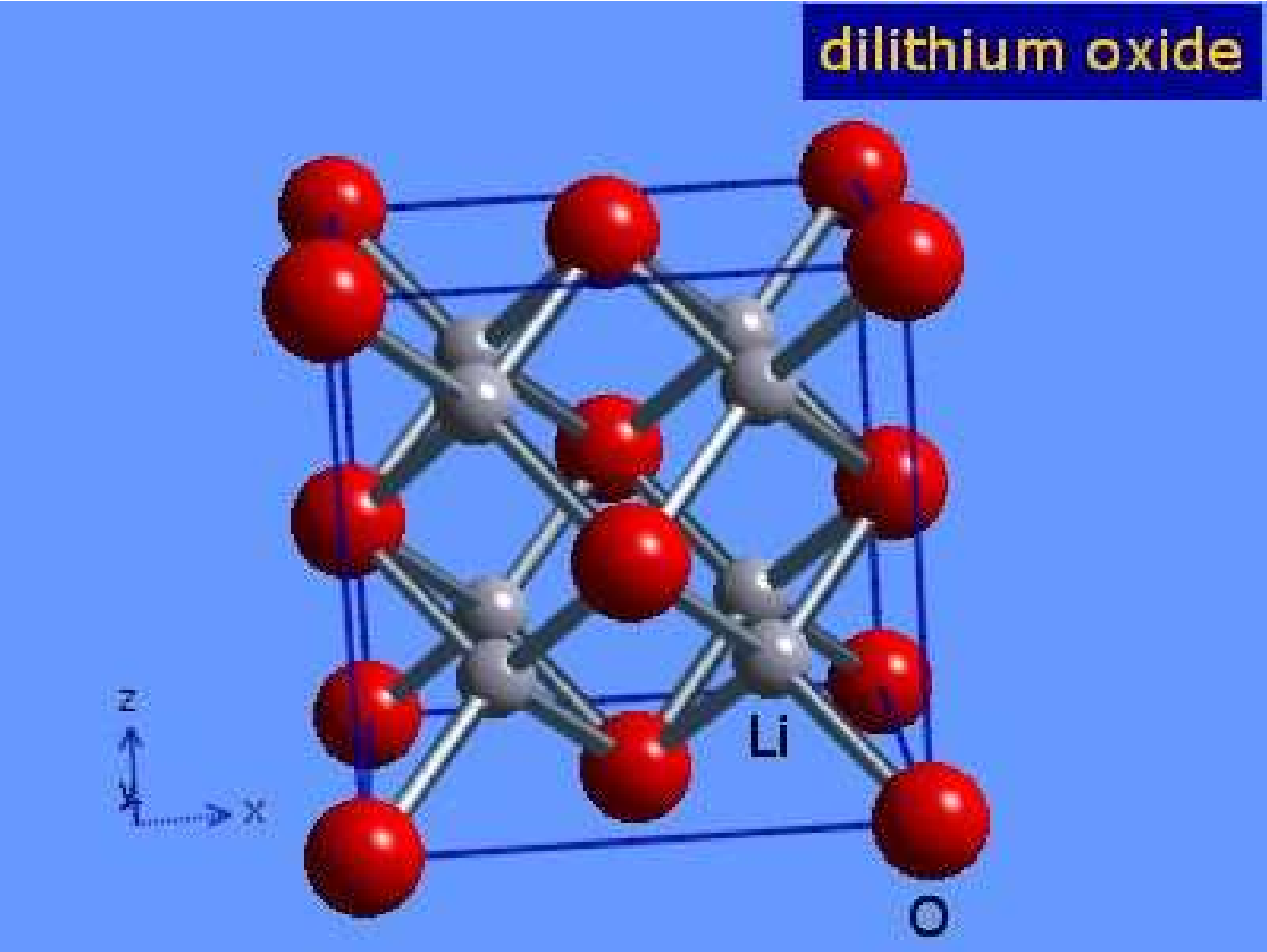
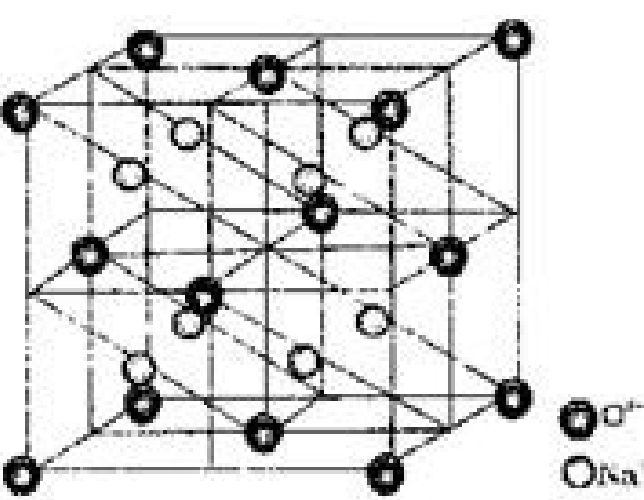


Fluorite



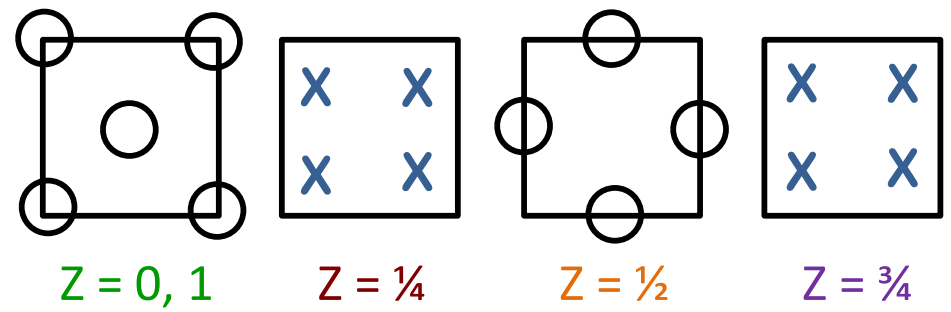
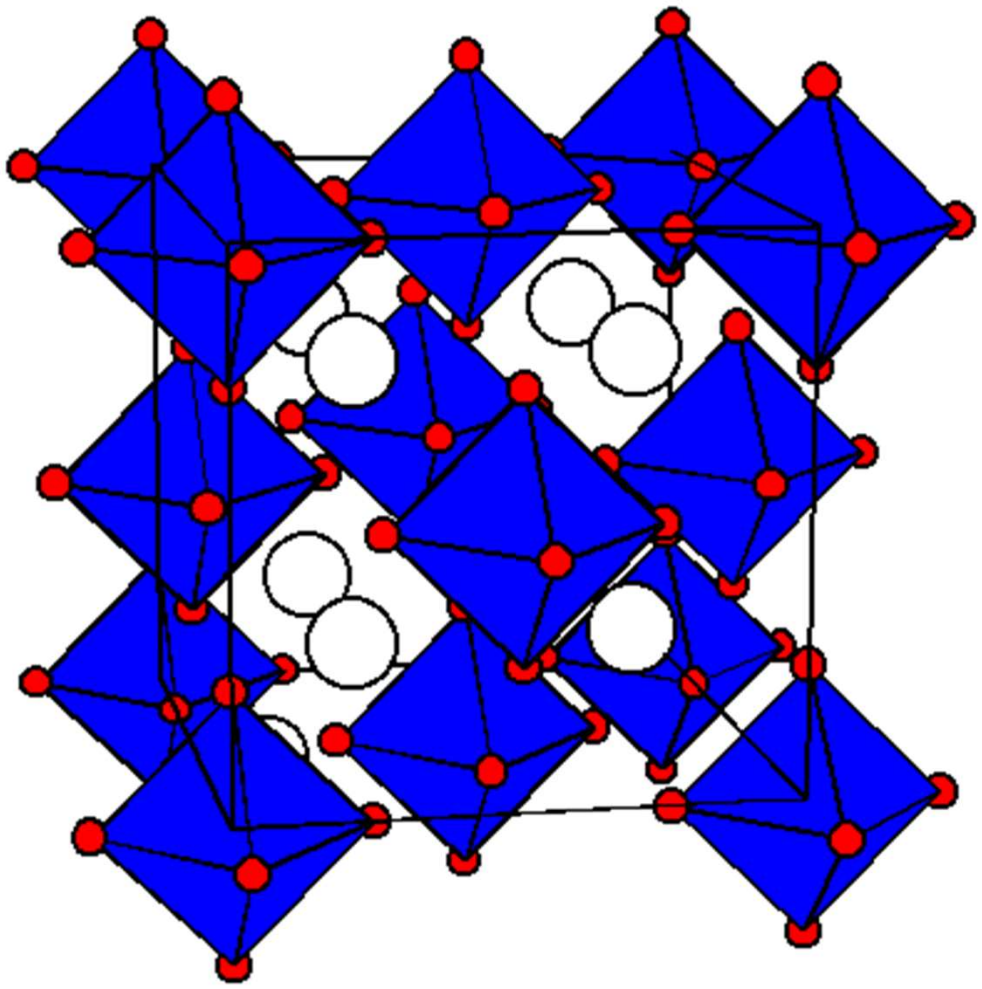
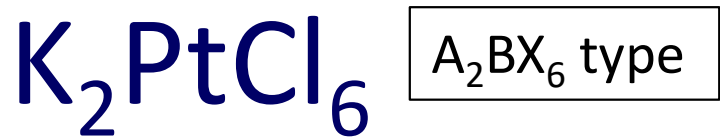
CaF_2 , SrF_2 , BaF_2 , SrCl_2 , BaCl_2 , CdF_2 , HgF_2 ,

Antifluorite



Li_2O , Li_2S , Li_2Se , Li_2Te , Na_2O , Na_2S , Na_2Se

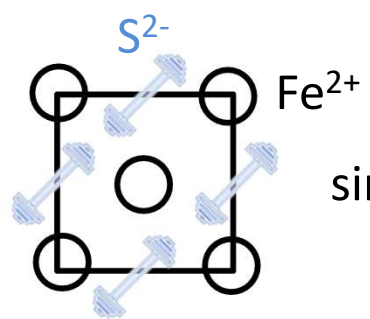
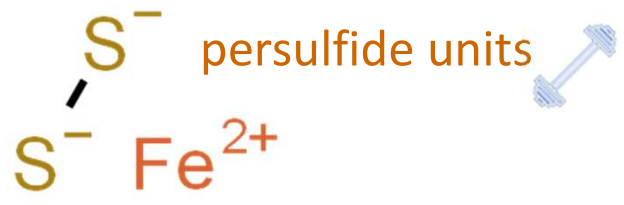
Complicated Antifluorite Structure



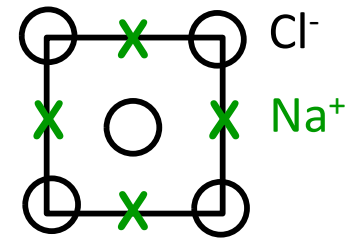
x K^+
o $PtCl_6^{2-}$

8 K^+ & 4 $PtCl_6^{2-}$

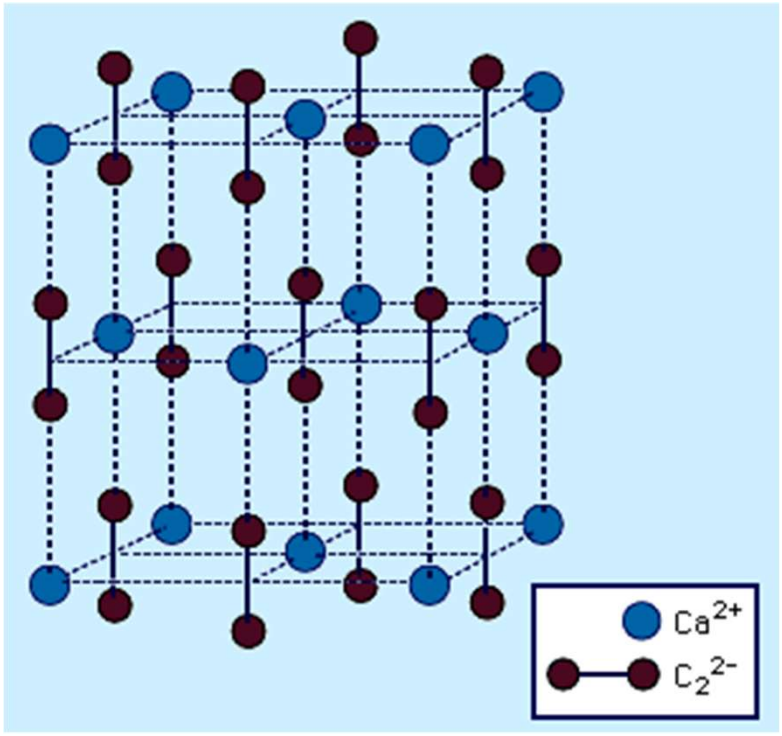
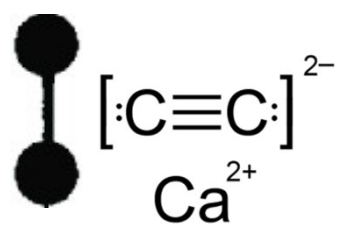
Complicated NaCl-type Structure



similar to



CdO_2 , PbSb_2 , PtSb_2



Cubic $\rightarrow$ Tetragonal

$$a' = \frac{a}{\sqrt{2}}$$

$$c \neq a$$

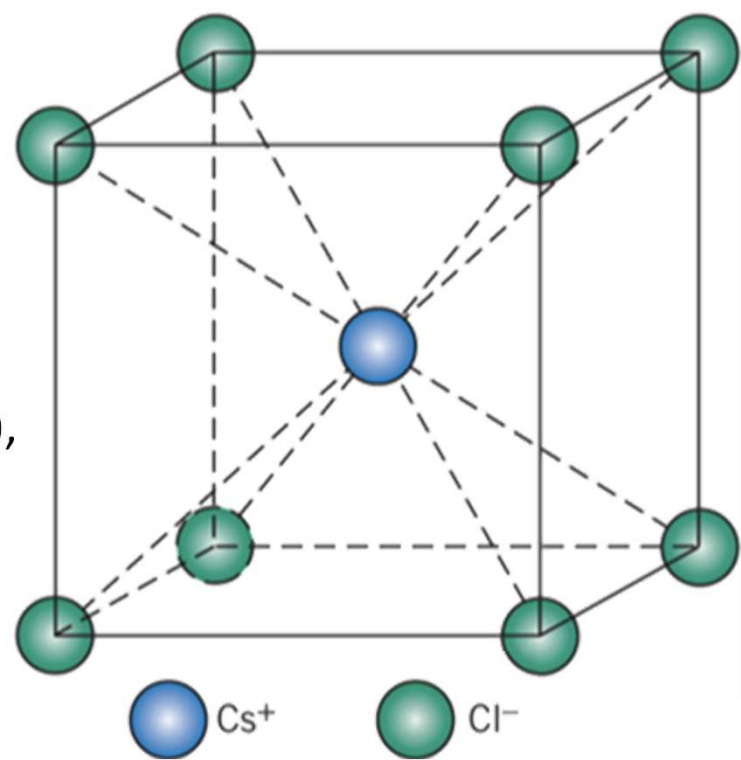
$$Ca \rightarrow 000 \quad 0\frac{1}{2}\frac{1}{2} \quad \frac{1}{2}0\frac{1}{2} \quad \frac{1}{2}\frac{1}{2}0$$

$$C \rightarrow 00z \quad 00\bar{z} \quad \frac{1}{2}\frac{1}{2}\frac{1}{2} + z \quad \frac{1}{2}\frac{1}{2}\frac{1}{2} - z$$

CaO_2 , BaO_2 , MoSi_2

Other Cubic Structures:

CsCl



β -brass (CuZn),
CsBr, CsI, AlN

A	B
000	$\frac{1}{2} \frac{1}{2} \frac{1}{2}$

In BCC, coordination is 8
In simple cube, coordination is 6

BCC lattice requires lattice points at the corners and at the center, both having the same motif

Here motif has changed!

Condition to get this structure:

(CN is 8) $\frac{R_A}{R_B} = \text{large}, > 0.73 \rightarrow \text{CsCl structure}$

same type of atom, CN is 6

different type of atoms, CN is 12

(CN is 6 or 12) $\frac{R_A}{R_B} = 0.41 - 0.73 \rightarrow \text{NaCl structure}$

TiO₂ (Rutile)

TiO₂ has 3 forms: Anatase, Brookite and Rutile

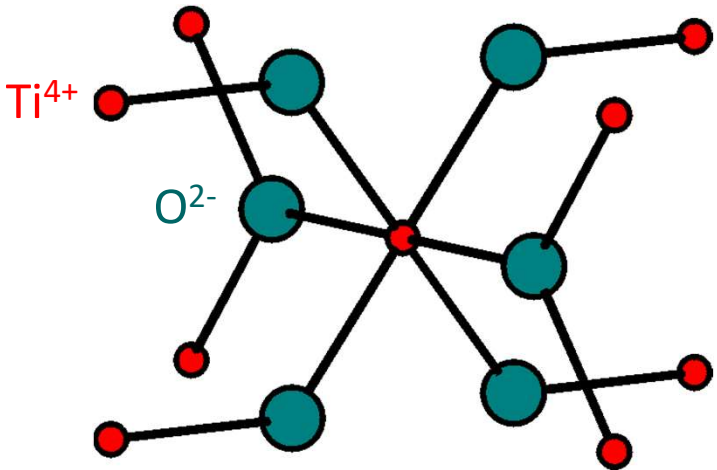
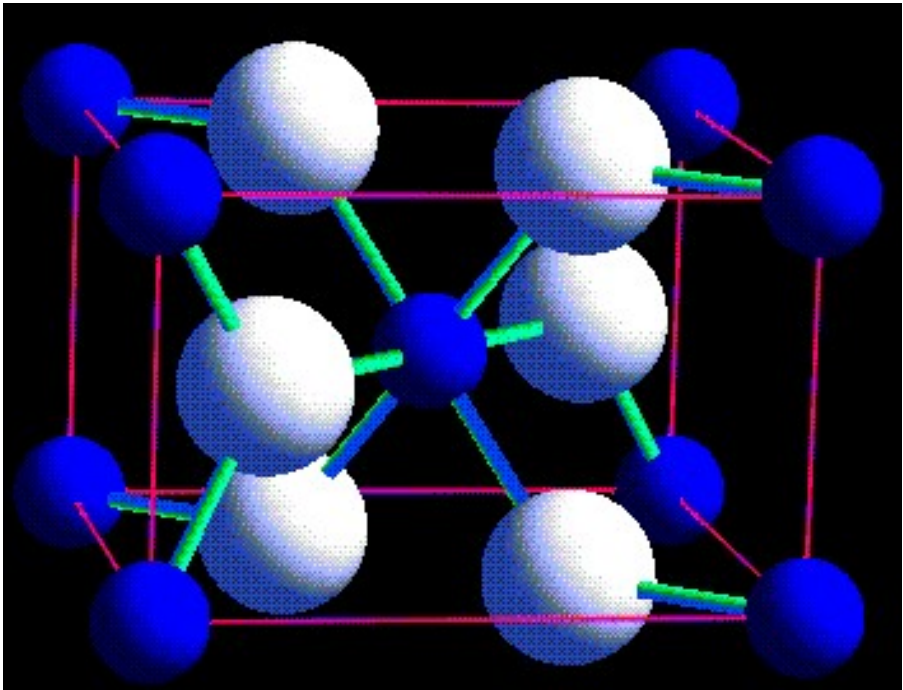
Rutile structure → radius ratio = 0.41-0.73

Each Ti⁴⁺ is octahedrally surrounded by 6 O²⁻ ions in a planar triangular arrangement

Coordination is 6 : 3

Rutile structure is not close packed. The unit cell is not a cube, one of the axes is 30% shorter than the other two.

The structure: **distorted BCC lattice of Ti⁴⁺ ions**



$$Ti \rightarrow 000 \quad \frac{1}{2} \frac{1}{2} \frac{1}{2}$$

$$O \rightarrow xx0 \quad \bar{x}\bar{x}0 \quad \frac{1}{2} + x, \frac{1}{2} - x, \frac{1}{2} \quad \frac{1}{2} - x, \frac{1}{2} + x, \frac{1}{2}$$

$$\frac{1}{4} \frac{1}{4} 0 \quad \frac{3}{4} \frac{3}{4} 0 \quad \frac{3}{4} \frac{1}{4} \frac{1}{2} \quad \frac{1}{4} \frac{3}{4} \frac{1}{2} \quad (\text{putting the value of } x = \frac{1}{4})$$

Actual value: $x = 0.3$

TiO₂ (Anatase)

For your knowledge only

